UM-1 analysis: the x0 mass mechanism and the dark sector of dirac16complex

dirac16complex project (with Claude Opus 5.5)

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Version 1 of 2026-10-17, written at the author's request of 2026-10-17: "analyze the x0 mass mechanism, and ... analyze the validity of associating dark matter (and dark energy) with dirac16complex". The model is defined in `MODEL.md` (version 1.1), its field equations are in `FIELD_EQUATIONS.md`, its approximate solution is in `SOLUTION.md`, and the origin of every ingredient is in `PROVENANCE.md`. This document examines two of the model's central claims against the model's own equations and numbers. The scripts are `analysis/x0_mass_analysis.py` (21 of 21 checks PASS, `analysis/reports/x0-mass-checks.json`) and `analysis/dark_validity.py` (13 of 13 checks PASS, `analysis/reports/dark-validity-checks.json`); every number below comes from them, and the tables of section 12 are generated by the second script. The labels are those of `MODEL.md`: DERIVED, TAKEN, STANDARD, ASSUMED, INVENTED, OPEN; ESTIMATE marks an order-of-magnitude calculation and FROM MEMORY a number quoted without a document at hand.

Coordinates are the author's: $X = \{x_0, x_1, \ldots, x_7\}$, x_0 the hidden space-like direction, $z = 6Hx_0$ real, $y = \ln|\sin z|/(6H) \in (-\infty, 0]$, the fold at $z = \pi/2$ ($y = 0$) and the tips at $z = 0, \pi$ ($y \to -\infty$, cut at $y = -L$ by the regular tip of the record). The block (x_0, x_5, x_6, x_7) has the scale factor $f = e^{-a_4} b_4^{1/2}$; $g_{00} = f^2 \cot^2 z$, so the proper hidden length is $ds = f\,|dy|$. The numbers use the vacuum of `SOLUTION.md`: $H = 23.42$ TeV, $\eta = HL = 3$, $f = 8^{-1/2}$, the proper half-cell $fL = 8.94 \times 10^{-21}$ m.

1. Summary: the answers

Part A, the x_0 mass mechanism.

1. What holds. Every four-dimensional mass of UM-1 is an eigenvalue of an x_0 operator acting on a wave function that is periodic in x_0 with the period $\pi/(6H)$, and the massless particles are exactly the graviton, the photon, the gluons and the fold-band quanta of Ψ. The author's requirement, mass from the x_0 dependence of the wave function, is met in this sense (DERIVED, section 3).

2. The exact mass formula (DERIVED, check A1_factorisation). In the flat proper measure every bosonic mass is $m^2 = \langle p^2 \rangle + (3H/2f)^2 - (3H/f)\,\rho(\text{fold}) + (3H/f)\,\rho(\text{tips}) + (\text{tip field})$, with ρ the normalised proper density. The uncertainty relation bounds the first term only. The second term is the warp of the three extra times, the third an ATTRACTIVE point potential at the fold, the fourth and fifth repulsive tip terms. Spin 1/2 has the same structure with the bulk mass in place of the warp (the Jackiw-Rebbi kink).

3. Where the Heisenberg argument holds: the x_0 excitations. Every odd mode and every excitation has $m \geq 3H/(2f) = 99.4$ TeV (the gap theorem, check A2_gap_theorem). For these modes the author's estimate $\hbar c/(2\Delta s)$ is a true lower bound, exceeded by factors 2.6 to 7.5. The dark-matter candidate of Ψ (50.9 TeV) belongs here: 78.9 % of its ε^2 is the x_0 standing wave. The Kohn-Sham energy-momentum tensor confirms this independently: the hidden pressure of the dark-matter quanta equals $-d\ln M_{DM}/d\ln f = 0.6575394$ to 1.1×10^{-9}.

4. Where it fails: the zero modes. The photon, gluon and graviton zero mode is localised at the fold within $\Delta s = 1.4 \times 10^{-21}$ m. The estimate gives it $\hbar c/(2\Delta s) = 70$ TeV, but its mass is exactly zero: the fold term cancels the hidden kinetic energy, and gauge invariance protects the cancellation. The same holds for the chiral fermion zero modes and the Ψ fold band. The

light masses of UM-1 are boundary effects at the tip. For the W, the tip term is 99.87 % of m_W^2. For the fermions, $m_f = m_{KK}\zeta(c)^2$ FALLS as the localisation sharpens, the reverse of the Heisenberg intuition. The ratio $m/(\hbar c/2\Delta s)$ is 1.1×10^{-3} for the W and 3×10^{-9} to 1.6×10^{-16} for e, u, d, ν.

5. Localisation in the coordinate x_0 is not physical. The zero mode, localised at the fold in proper length, has 62 % of its probability within $\pi/8$ of the tips in the coordinate x_0, because $g_{00} = f^2 \cot^2 z$ vanishes at the fold. Only the proper form $\Delta s\, \Delta p \geq \hbar/2$ says anything about masses, and on a compact period not even that in general (the Ψ fold band has $\langle p^2 \rangle \Delta s^2 = 0.20 < 1/4$).

6. Every x_0-generated mass tracks the block size f: $d \ln m / d \ln f = -1$ (KK modes), -0.50 (W, Z), -20 to -39 (e, u, d, ν), -0.66 (dark matter); α and G_N go like f^{-4}, and m_p/m_e like f^{-29} (ESTIMATE). Atomic clocks require $|h_4|/H_0 < 1.2 \times 10^{-8}$ today (FROM MEMORY); UM-1 has 10^{-93}. The primordial deflation $h_4 = -H/3$ would change m_e by a factor e every 4×10^{-30} s, so it must end long before nucleosynthesis.

7. Verdict on the mechanism. The mechanism is VALID as a kinematic statement (all masses are x_0 eigenvalues of periodic wave functions). It is dynamically correct, Heisenberg estimate included, for the heavy sector. It is NOT the origin of the light Standard-Model masses: taken literally as a principle, it would put every massive particle above the gap of 99 TeV.

Part B, the dark sector.

1. Dark matter: the association is CONDITIONAL. The heavy odd quanta of Ψ ($M_{DM} = 50.9$ TeV) are cold on the static block ($w = 0$), dark (a gauge singlet), collisionless ($\sigma/M \sim 3 \times 10^{-18}$ cm^2/g) and get their mass from x_0. Not established:

   - stability: it needs the fold parity exact to 2×10^{-27} (ASSUMED);
   - the relic abundance: it needs $\langle \sigma v \rangle = 6.5 \times 10^{-26}$ cm^3/s, 37 % of the s-wave unitarity limit, i.e. a contact strength $G \approx 1.4/H^2$ of natural size, not derived (OPEN);
   - the computed Kohn-Sham ground states are dark RADIATION, not dark matter;
   - while the block deflates, the quanta have $w_{eff} = -0.073$, not 0;
   - TENSION: the portal that gives $N_{eff} = 2.99$ (decoupling of the fold band in the QCD crossover), applied to the dark matter with its x_0 overlap, makes it under-abundant ($\Omega h^2 = 0.023$) and puts σ_{SI} at 1.6 times the LZ limit.

   A thermal relic through the portal alone is consistent ($G^{-1/2} = 32.5$ TeV, $\sigma_{SI} = 8.8 \times 10^{-46}$ cm^2) and testable.

2. Dark energy: the association is NOT SUPPORTED. No Ψ state of positive energy has $w < 0$ on the static block. The $\lambda < 0$ condensate tuned to $w_0 = -1$ has $w_a = -6$, and its energy density turns negative at $z = 0.26$. A natural vacuum energy of Ψ exceeds the dark energy by 6×10^{64}. The dark energy of UM-1 is $\delta\Lambda$, a tuned geometric cosmological constant.

3. Consequences for the documents (section 10). Statements of `MODEL.md` (sections 1 and 4.4), `FIELD_EQUATIONS.md` (check D3) and `SOLUTION.md` (section 5.2) that said the uncertainty relation bounds every mass, or quoted the Standard-Model relic value, are corrected and point here.

2. The question, the definitions and the method

The author's argument (`PROVENANCE.md` section 1) is that localisation in x_0 implies, through $\Delta x_0\,\Delta p_{x_0} \geq \hbar/2$, a momentum along x_0. That momentum is proportional to a mass, so every particle whose wave function depends periodically on x_0 acquires an effective mass, except the graviton, the photon and perhaps the quanta of the dirac16complex field. Three definitions make the argument testable.

- The momentum that is a mass. For an eight-dimensional massless field the four-dimensional mass is $m^2 = g^{00} p_0 p_0 = \hat{p}^2$, the square of the hidden momentum in the orthonormal frame, $\hat{p} = -i\hbar\,\partial_s$ with s the proper hidden length. The coordinate momentum p_{x_0} differs from $\hat{p}$ by the position-dependent factor $f|\cot z|$, which vanishes at the fold.
- The mass of a mode is not a local momentum but an eigenvalue: m^2 is the eigenvalue of the x_0 operator of the field with the fold and tip conditions (section 3). The question is whether that eigenvalue is bounded below by the hidden kinetic energy $\langle \hat{p}^2 \rangle \geq \hbar^2/(4\Delta s^2)$.
- The localisation Δs is the standard deviation of the proper density on one full period (both halves of the cell, the fold in the middle). `SOLUTION.md` used one half, which gives a Δs smaller by about $\sqrt{2}$; no conclusion depends on the choice.

The analysis is exact where it can be (sympy, part A1 of the script) and numerical otherwise. Part A2 uses two independent finite-element discretisations of the same eigenproblem, compared with the exact spectra. Part B evaluates the standard criteria for dark matter and dark energy with the numbers of the solution.

3. The x_0 operators of all fields (DERIVED, checks A1)

3.1 Scalars, gauge fields and the graviton

For the corrected metric (version 2), $\sqrt{|g|} = f^4 e^{3a_4} |\cot z|\,|\sin z|^{1/2}$, so that $\sqrt{|g|}\,dx_0 = e^{3a_4} f^4 e^{3Hy}\,dy$. The x_0 part of the d'Alembertian of a field that depends on x_0 through u is (check A1_scalar_x0_operator, exact for a generic profile)

$$-\frac{1}{f^2}\, e^{-3Hy}\, \partial_y (e^{3Hy}\, \partial_y u) = m^2\, u.$$

The four-dimensional components of a gauge field and the transverse-traceless graviton see the same operator, because the four-dimensional part of the metric does not depend on x_0. For the graviton this is exact in Einstein gravity. With the Lovelock terms the zero mode stays massless by four-dimensional general covariance, and the excited spectrum carries background-curvature factors that are not computed here (OPEN).

- Fold conditions (check A1_fold_conditions_from_flux). The flux of the operator through x_0 is $e^{3a_4} f^2 \,\mathrm{sign}(\cot z)\, e^{3Hy} u_y$. Continuity across the fold gives $u_y(0) = 0$ for a mode EVEN under the fold reflection $z \to \pi - z$ (Neumann), and $u(0) = 0$ for an ODD mode (Dirichlet).

- The flat form (check A1_flat_form_warp_potential). With $u = e^{-3Hy/2}v$ the measure becomes flat and the operator becomes $f^{-2}(-v'' + \frac{9}{4}H^2 v)$: the warp of the three extra times is a constant potential $(3H/2f)^2$, and the Neumann condition becomes the Robin condition $v'(0) = \frac{3}{2}Hv(0)$. On one full period in the proper coordinate $s \in [-fL, fL]$ (fold at $s = 0$, tips at $s = \pm fL$, identified by the periodicity) this is the Schroedinger problem

$$-\partial_s^2 v + V(s)\, v = m^2 v, \qquad V(s) = \Big(\frac{3H}{2f}\Big)^2 - \frac{3H}{f}\,\delta(s) + \frac{3H}{f}\,[\delta(s - fL) + \delta(s + fL)] + \beta\text{-terms of the tip field.}$$

- The factorisation (check A1_factorisation). With $\phi = \frac{3H}{2f}|s|$, the operator is $A^\dagger A$ with $A = \partial_s + \phi'$, the structure of supersymmetric quantum mechanics. So

$$m^2 = \frac{\int ds\, |(\partial_s + \phi')v|^2}{\int ds\, |v|^2} + (\text{tip field}) = \frac{\int dy\, e^{3Hy} |\partial_s u|^2}{\int dy\, e^{3Hy} |u|^2} + (\text{tip field}).$$

This is the precise form of the author's idea. The mass is the x_0 dependence of u, the profile measured RELATIVE to the geometric zero mode $v_0 = e^{-\phi}$. A mode localised exactly as the geometry dictates ($u = \text{const}$) is massless. Every deviation from it costs mass.

- Without a cutoff (check A1_exact_spectrum_without_cutoff). For $L \to \infty$ the only normalisable mode below $3H/(2f)$ is $u = e^{(\kappa - 3H/2)y}$. The fold condition forces $\kappa = 3H/2$, i.e. $m = 0$ (the flat zero mode, of finite weighted norm $1/(3H)$), and the odd condition forbids it. Above $3H/(2f)$ the spectrum is continuous. The proper length of a half cell is infinite, but its weighted volume is finite, so four-dimensional gravity and the gauge couplings are finite without the cutoff.

3.2 Spin 1/2 (TAKEN block, DERIVED identity)

The Kohn-Sham record reduces Ψ and every Standard-Model fermion to the 2×2 block $h = -i\sigma_1\partial_t + c\,\sigma_2$ ($t = y/L$, $c = f\mu L$, units $1/(fL)$), in which the spin connection removes the warp. For every profile χ (check A1_spin_half_boundary_identity)

$$\int |h\chi|^2 = \int |\chi'|^2 + c^2 \int |\chi|^2 - c\,[\chi^\dagger \sigma_3 \chi]_{-1}^{0}.$$

The chiral zero mode $(e^{ct}, 0)$ is annihilated by h. Its kinetic energy c^2 and bulk mass c^2 are cancelled by the fold term $-2c^2/(1 - e^{-2c})$: the Jackiw-Rebbi mechanism of a kink in the bulk mass. It is localised within $1/(2c)$ yet massless.

3.3 The tip parameter

Matching the bulk gauge kinetic term $\hat{g}^{-2} e^{3a_4} f^2 e^{3Hy} (A_y)^2$ to the tip mass term of the sigma field, with $\sqrt{|g_{tip}|} = e^{3a_4} f^3 e^{3Hy}$, gives the Robin condition $u'(-1) = \beta u(-1)$ with $\beta = c\,\hat{g}^2 v_\Sigma^2 fL$ (check A1_tip_parameter_scaling). So β is proportional to f, and the warp factors cancel. The docstring of `solution/masses.py` writes $c_{tip} f^2 L e^{3\eta}$, a different parametrisation of the same fitted β. The fit is unaffected, and only the f dependence of section 7 uses the scaling derived here.

3.4 What the period $\pi/(6H)$ means

Without a cutoff, every period of x_0 is a whole real line in proper length, and consecutive periods are separated by tips at infinite proper distance. Every function of y is automatically periodic, so periodicity alone constrains nothing. With the regular tip of the record a period is a circle of proper circumference $2fL$. Its modes are classified by the two reflections, about the fold and about the tip. The content of the author's requirement lies in these parities and in the boundary terms at the fold and the tips.

4. The spectra and the gap (checks A2)

- Two discretisations, one problem (check A2_spectra_two_discretisations). The flat v-form and the weighted u-form finite elements (600 and 1200 elements, Richardson) reproduce the exact levels of the four boundary classes to 2.6×10^{-9}. The classes are: fold even or odd, tip Neumann or Dirichlet; their exact formulas are in the script. In units $(mfL)^2$, with the gap at $9\eta^2/4 = 20.25$:
  - even and Neumann tip: $0, 30.12, 59.73$;
  - even and Dirichlet tip: $0.01002, 35.07$ (the largest W a positive tip term can give);
  - odd and Neumann tip: $27.09, 49.48$;
  - odd and Dirichlet tip: $30.12, 59.73$.
- The gap theorem (check A2_gap_theorem). For tip parameters β from 0 to 10^6, the even sector has exactly one level below the gap $(3H/2f)^2$ (the zero mode, lifted only by the tip) and the odd sector has none. Every odd mode and every excitation has $m \geq 3H/(2f) = 99.4$ TeV. The reason: the Dirichlet condition removes the attractive fold term, and a rank-one attraction binds at most one level.
- Without the cutoff (check A2_no_cutoff_limit). For $\eta = 3, 6, 12, 24$ the zero mode stays massless, and the first even excitation falls to the gap from above: $m_1 f/H = 1.8294, 1.5888, 1.5227, 1.5057$.

5. The mass of every particle, term by term (checks A2)

The table of section 12.1 splits $(mfL)^2$ of every particle of UM-1 into the hidden kinetic energy $\langle p^2 \rangle$, the warp (or bulk mass), the fold term and the tip terms. It compares the mass with the author's estimate $m_H = \hbar c/(2\Delta s)$. The sums reproduce the eigenvalues to 2.8×10^{-4} (check A2_decomposition_sum_rule).

- Photon, gluons, graviton (check A2_localised_massless_modes). $\Delta s = 0.157\, fL = 1.4 \times 10^{-21}$ m and $\langle p^2 \rangle = 20.25 = (3H/2f)^2 (fL)^2$, so $m_H = 70.5$ TeV. Yet $m = 0$: warp $+20.25$ and fold -40.50 cancel the kinetic energy exactly. The uncertainty relation holds ($\langle p^2 \rangle \Delta s^2 = 0.497$) but does not bound the mass.

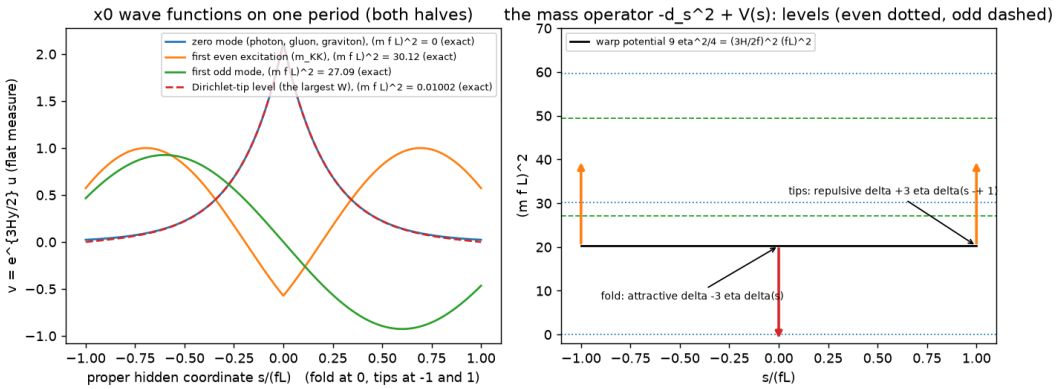


Figure 1: The x0 wave functions on one period (flat measure) and the mass operator: warp potential, attractive fold, repulsive tips, the levels.

- W and Z (check A2_W_mass_is_a_tip_term). In the dressed form, the bending of the profile is 1.32×10^{-3} of m_W^2 and the sigma-field tip term is 0.99868. In the flat form the W has the same kinetic energy as the photon, 1.5×10^6 times m_W^2. $m_W/m_H = 1.1 \times 10^{-3}$.

- Kaluza-Klein and odd modes (check A2_heisenberg_holds_for_excitations): $m/m_H = 7.49$ (first even) and 6.47 (first odd). The odd mode has no fold term.

- Fermions (check A2_localisation_lowers_fermion_masses). Each zero mode has kinetic c^2, bulk c^2 and fold $-2c^2$ (sum 0 to 2.4×10^{-5}). Its mass $m_f = m_{KK}\zeta(c)^2 \approx 2c\,e^{-2c} m_{KK}$ comes from the tip Yukawa coupling. The electron has $c = 11.20$, $\Delta s = 5.6 \times 10^{-22}$ m and $m_H = 175$ TeV, while $m_e = 0.511$ MeV. A sharper localisation (larger c) gives a LIGHTER fermion: $d\ln m_f / d\ln c = 1 - 2c - 2c\,e^{-2c}/(1 - e^{-2c})$.

- Ψ (check A2_dark_matter_mass_is_x0_momentum). The fold band is a chiral zero mode ($c = fmL = 1.061$), localised but massless. The lowest odd level satisfies $a(0) = 0$, $b(-L) = 0$ and $\tan q = -q/c$, which gives $q = 2.0486$ and $\varepsilon = 2.174920403411\,m$, equal to the record to 10^{-12}. The x_0 standing wave carries 78.9 % of ε^2, and its fold and tip terms are POSITIVE, so $m^2 \geq \langle p^2 \rangle + m_{bulk}^2$ and $m/m_H = 2.61$. The lowest even bulk level ($3.1262\,m$) has $m/m_H = 4.07$.

6. The uncertainty relation: what it bounds

- The proper uncertainty relation $\Delta s\,\Delta\hat{p} \geq \hbar/2$ bounds $\langle \hat{p}^2 \rangle$, the first term of $m^2 = \langle \hat{p}^2 \rangle + \langle V \rangle$. It bounds the mass only when $\langle V \rangle$ is at least the warp gap, i.e. when the fold term is absent or repulsive. That is the case for every odd mode, every excitation and the Ψ bulk band.

- For the zero modes $\langle V \rangle = -\langle \hat{p}^2 \rangle$ exactly. This is not a numerical accident. The cancellation is protected by four-dimensional gauge invariance (photon, gluons), general covariance (graviton) and chirality, i.e. the index of the Dirac block (the fermion zero modes and the Ψ fold band). The author's exception list (graviton, photon, perhaps the Ψ quanta) is therefore exactly the list of protected zero modes, extended by the gluons.



Figure 2: The terms of (m f L)^2 for every particle: the zero modes are localised but massless because the fold term cancels.

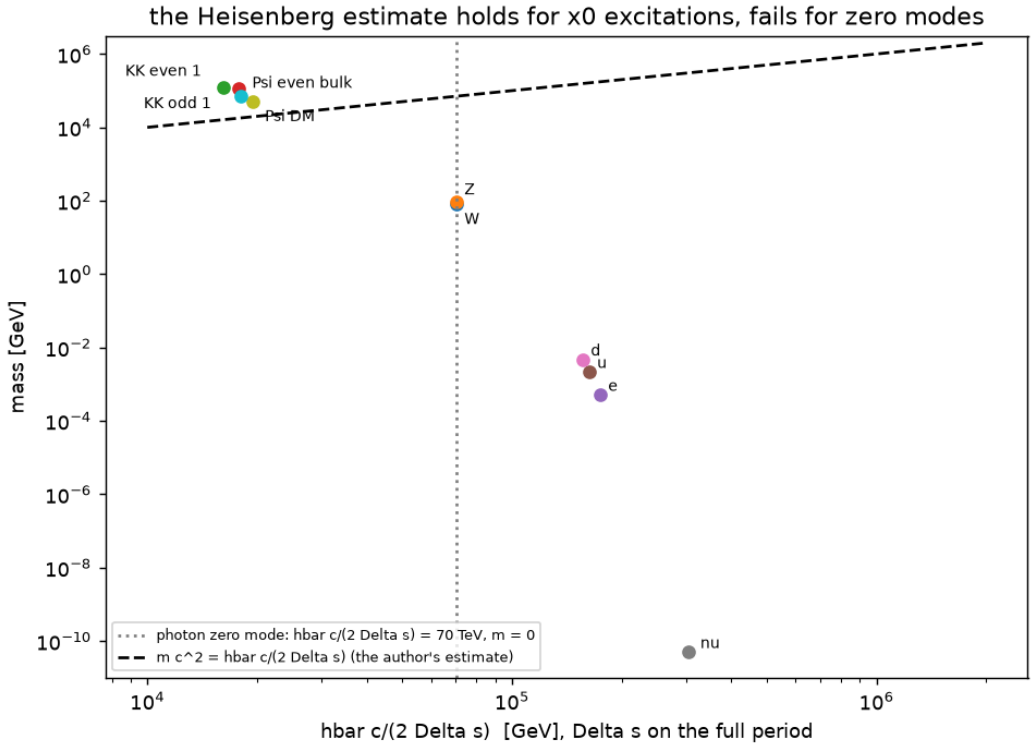


Figure 3: The actual mass against the author's estimate hbar c/(2 Delta s): the x0 excitations lie above the line, the zero modes far below it.

- Localisation in the coordinate x_0 (check A2_coordinate_localisation_is_not_physical, figure below) is chart-dependent. Since $g_{00} = f^2 \cot^2 z$ vanishes at the fold, a short proper interval there spans a long coordinate interval. The zero mode that sits at the fold in proper length has 62 % of its probability within $\pi/8$ of the tips in x_0 ($\Delta x_0/\ell_0 = 0.40$). The coordinate relation $\Delta x_0\, \Delta p_{x_0} \geq \hbar/2$ therefore says nothing about masses.
- On a compact period the position is not a canonical observable. The uncertainty product can fall below 1/4: the Ψ fold band has $\langle p^2 \rangle \Delta s^2 = 0.20$, because its density at the tips is 0.12 of its fold value. For a compact cell the correct statement is the spectral gap of section 4, not the uncertainty relation.

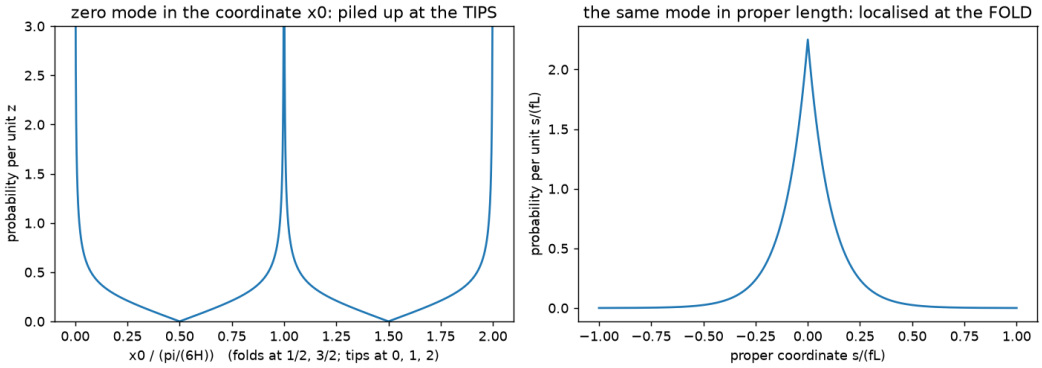


Figure 4: The zero mode over two periods: piled up at the tips in the coordinate x0 (left), localised at the fold in proper length (right).

7. The block size and the constancy of the masses

At fixed eight-dimensional parameters every x_0-generated mass depends on f (checks A2_sensitivities and A2_deflation_bound_today; table of section 12.2):

- the Kaluza-Klein masses and the gap go like $1/f$;
- W and Z go like $f^{-0.50}$ ($\beta \propto f$, $d\ln\Lambda/d\ln\beta = 0.9987$);
- the fermions go like f^{-2c}: -22.4 (e), -20.9 (u), -20.1 (d), -39.1 (ν), with closed forms agreeing to 10^{-7};
- the dark matter goes like $f^{-0.658}$;
- α and G_N go like f^{-4} (the proper volume $V_\perp \propto f^4$);
- Λ_{QCD} goes like $f^{-51.8}$ (one loop, $n_f = 2$, ESTIMATE), so m_p/m_e goes like $f^{-29.4}$.

The atomic-clock limits on $\dot\alpha/\alpha$ and $\dot\mu/\mu$ (FROM MEMORY: Lange et al. 2021, PRL 126, 011102, 2-sigma) give $|h_4|/H_0 < 1.2 \times 10^{-8}$ today; UM-1 has $h_4/A = -8.9 \times 10^{-94}$ (`SOLUTION.md` section 3.4). The record's primordial deflation $h_4 = -H/3$ would change m_e by a factor e every 3.8×10^{-30}

s. The exponential deflation of the extra times must therefore end (the static block of UM-1) long before nucleosynthesis; after that, the block deflates only slowly and never stops.

The hidden-pressure identity (check A2_hidden_pressure_is_the_x0_mass). The pressure of a quantum on the hidden direction is $-dE/d\ln(\text{proper length})$. The Kohn-Sham energy-momentum tensor of eight dark-matter quanta, solved self-consistently on three grids (`solution/outputs/ks/dm_lam0.json`), gives $p_0/\rho = 0.6575394020$ and $p_t = 0$. The x_0 eigenvalue gives $-d\ln\varepsilon/d\ln f = 0.6575394031$. Two independent computations, one from the stress tensor and one from the spectrum, agree to 1.1×10^{-9}. The ratio would be 1 for a mass that is pure x_0 momentum and 0 for a bulk mass.

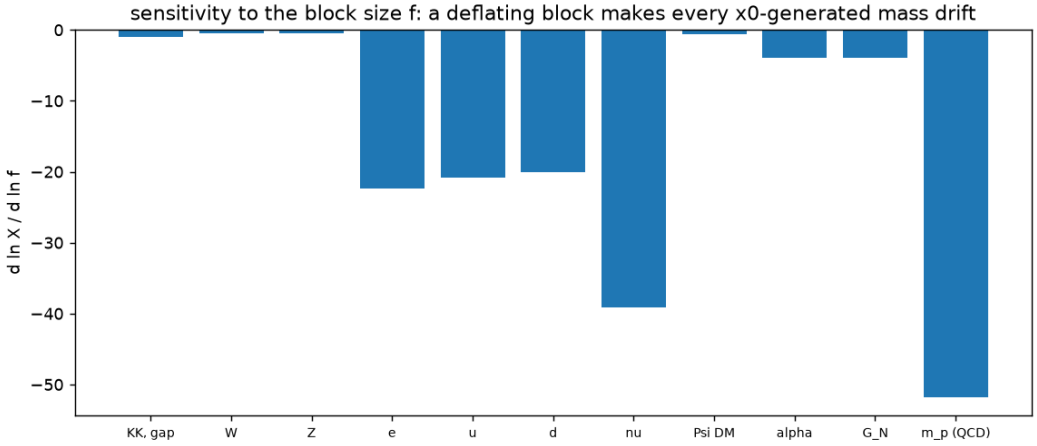


Figure 5: Sensitivity of every x0-generated quantity to the block size f.

8. Is dirac16complex the dark matter? (checks B)

8.1 Which quanta

Ψ has three kinds of quanta in UM-1 (`MODEL.md` section 4.2).

- The fold band: massless chiral zero modes, $w = 1/3$.
- The bulk band: massive, with the lightest odd level at $2.17492\,m = 50.9$ TeV.
- The homogeneous condensate.

The dark-matter hypothesis can only refer to the bulk band. The Kohn-Sham ground states computed in the record and in UM-1 fill the fold band exactly (check B6_the_computed_ground_states_are_radiation). They are dark RADIATION ($\Delta N_{eff} = 1.98$ in the N_{eff} fit), and they do not populate the bulk band.

8.2 The criteria

criterion	**result**	**status**
cold	$w = 0$ ($\lambda = 0$), 0.0018 (λ_1) on the static block (B1)	PASS
dust in the primordial era	$w_{eff}(A) = -0.0731$ while the block deflates, from p_0 and from the x_0 mass alike (B1)	not dust before the block is static
massive, mass from x_0	50.9 TeV, 78.9 % x_0 momentum, p_0/ρ identity (A2)	PASS
dark	gauge singlet; the dipole portal P3 is unbounded here	ASSUMED, P3 OPEN
stable	lightest odd state; a parity violation must stay below 1.8×10^{-27} (B1)	ASSUMED
relic abundance	6.5×10^{-26} cm^3/s needed, 37 % of unitarity, $G \approx 1.42/H^2$ (B3)	OPEN (natural size)
consistent with the N_{eff} portal	$\Omega h^2 = 0.023$, $\sigma_{SI} = 1.6\times$ LZ limit (B4)	TENSION
relic through the portal alone	$G^{-1/2} = 32.5$ TeV, $\sigma_{SI} = 8.8 \times 10^{-46}$ cm^2 (B4)	consistent, testable
collisionless	$\sigma/M = 2.9 \times 10^{-18}$ cm^2/g for $G = 1/H^2$ (B5)	PASS
Pauli (Tremaine-Gunn)	50.9 TeV against a bound of order keV	PASS
back-reaction on the block	$\delta k/k_0 = 8 \times 10^{-95}$ today (B7)	PASS

8.3 Stability

The lightest odd level ($2.174920\,m$) lies below the lightest even bulk level ($3.126177\,m$). With exact fold parity it cannot decay. If a tip term or a portal breaks the parity with an amplitude ϵ_P, the width is of order $\epsilon_P^2 M/(8\pi)$. A lifetime above 10^{26} s (the order of the indirect-detection limits) then requires $\epsilon_P < 1.8 \times 10^{-27}$. The parity must be an exact symmetry of the full model, including the tip sigma field and the portals at both tips. UM-1 assumes it; it is not derived.

8.4 The relic abundance

The vector-contact annihilation is computed by an explicit Dirac trace (check B2_vector_contact_annihilation): $\sum|\mathcal{M}|^2 = 128\,G^2M^4$ at threshold, so $\sigma v_{rel} = G^2M^2/\pi$ per massless Dirac final state (half for a chiral one). Freeze-out (Kolb-Turner, STANDARD) with the light states of UM-1 ($g_* = 60.25$: 53.25 of the one-generation Standard Model above m_W, plus the fold band) gives $x_f = 29.5$. It requires $\langle\sigma v\rangle = 6.47 \times 10^{-26}$ cm^3/s. The familiar 4.4×10^{-26} assumes $g_* = 106.75$; rescaled it is 6.9×10^{-26}. This is 37 % of the thermally averaged s-wave unitarity limit 1.73×10^{-25} cm^3/s, which a 50.9 TeV relic would reach at 83 TeV. As one contact

channel, the required strength is $G^{-1/2} = 19.6$ TeV, i.e. $G = 1.42/H^2$: the natural size of an interaction of the cell. Whether $\lambda(\bar{\Psi}\Psi)^2$ (whose scalar-scalar part annihilates only in the p-wave) and the portals supply it is OPEN.

8.5 The tension with the N_{eff} portal

`SOLUTION.md` fits $N_{eff} = 2.99$ with the fold band decoupling in the QCD crossover. A neutrino-like estimate gives the corresponding vector portal $G^{-1/2} = 12.3$ TeV (ESTIMATE). The four-dimensional portal coupling of a Ψ mode is proportional to its overlap with the fold-localised Standard-Model fermions (figure B2). The dark matter's overlap is 0.329 of the fold band's, so its portal is $G^{-1/2} = 21.5$ TeV. Annihilation into e, ν, u, d (7.5 Dirac-equivalent channels) then gives 5.2 times the relic cross-section, i.e. $\Omega h^2 = 0.023$: UNDER-abundant. Coherent scattering on nucleons gives $\sigma_{SI} = 4.6 \times 10^{-45}$ cm^2. That is 1.6 times the LZ limit 2.8×10^{-45} cm^2 at 50.9 TeV, linearly extrapolated from the published minimum (2.2×10^{-48} cm^2 at 40 GeV, PRL 135, 011802 (2025)). The extrapolation is weaker than the published curve, so the tension is conservative.

The dark-matter reading of the bulk band and the N_{eff} reading of the fold band are therefore in tension unless the portal coefficients of the two modes differ by an $O(1)$ factor beyond their overlaps, or the one-half count ($g_F = 4$) of `SOLUTION.md` holds. Conversely, a relic that annihilates through the vector portal alone needs $G^{-1/2} = 32.5$ TeV and predicts $\sigma_{SI} = 8.8 \times 10^{-46}$ cm^2 (0.31 of the limit), within reach of the next xenon experiments.

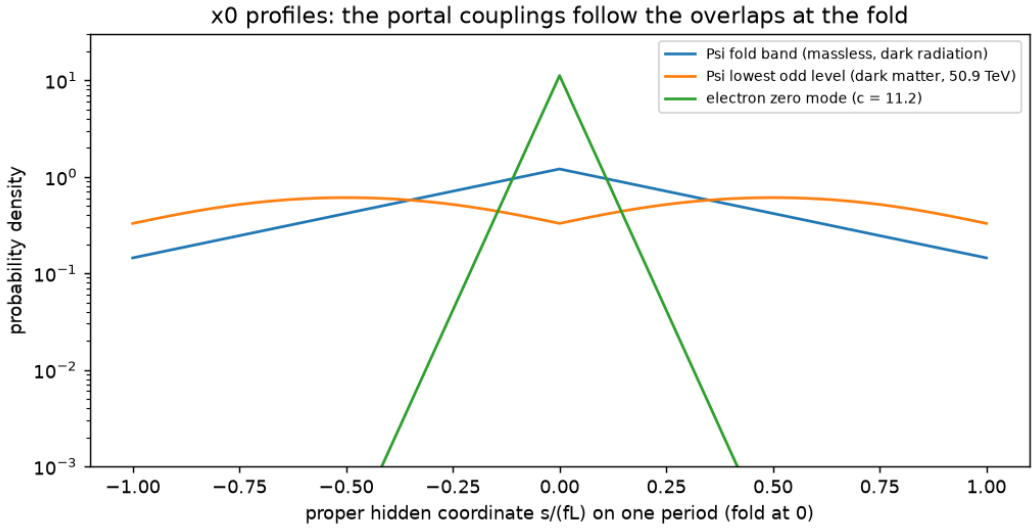


Figure 6: The x0 profiles of the Psi fold band, the Psi dark matter and the electron: the portal couplings follow the overlaps at the fold.

8.6 The primordial era

On the record's history ($\beta = 4/3$) the bulk quanta are blueshifted by the shrinking hidden direction: $w_{eff}(A) = (\beta/2 - 1)p_0/(3\rho) = -0.0731$, equal to $-\frac{1}{3}\, d\ln\varepsilon/da_4$ from the x_0 mass (check B1_not_dust_while_the_block_deflates). This is the record's finding that the bulk band falls

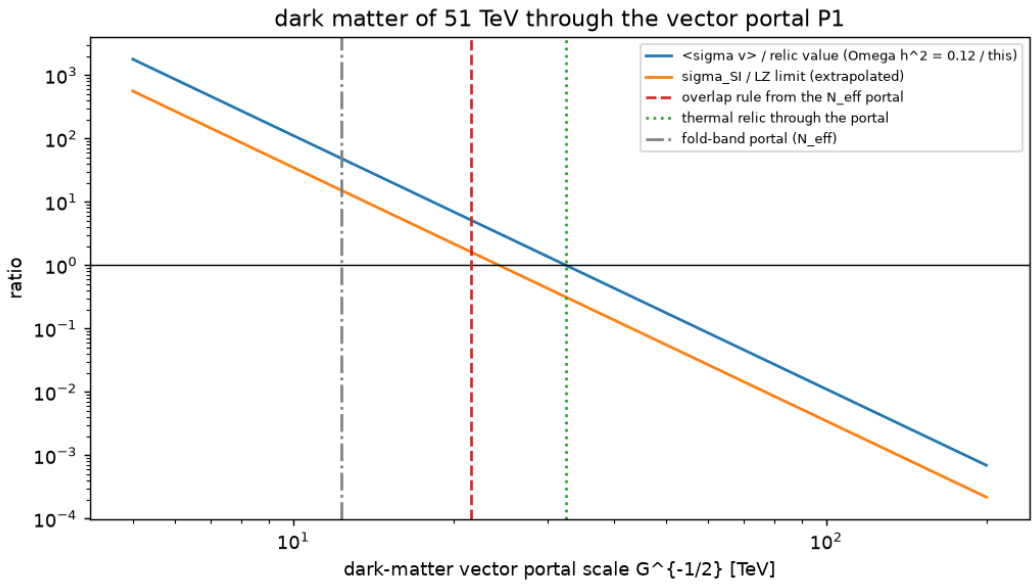


Figure 7: Dark matter of 51 TeV through the vector portal: relic and direct-detection ratios against the portal scale.

through $w = 0$ (`Revision/dark_sector/dirac16complex/README.md`). The dark matter is dust only once the block is static.

8.7 Verdict

CONDITIONAL. The heavy odd quanta of Ψ are a valid cold-dark-matter candidate of UM-1 provided three things hold: the fold parity is exact, an annihilation of natural cell strength exists, and the portal coefficients are not universal across modes. The dirac16complex field as a whole is NOT "the dark matter": its computed states are dark radiation, and its dark-matter sector is a heavy Kaluza-Klein-like excitation whose abundance the model does not yet predict.

9. Is dirac16complex the dark energy? (checks E)

- Positive-energy states (check E2_no_positive_energy_state_accelerates). On the static block the record's shift $(\beta/2 - 1)(P_0 + 3P_t)/(3E)$ vanishes, so $w_{eff} = P_3/E \geq 0$ for every real-frequency state. The values are: dark-matter quanta 0 and 0.0018, fold band 1/3, the record's interacting gas 0.3306 to 0.3361. None accelerates the expansion. On the primordial history the record's bound $w_{eff}(C) \geq -4/9$ applies (TAKEN).
- The condensate (check E1_condensate_cannot_be_dark_energy, table of section 12.4). On the static block $S = \bar{\Psi}\Psi \propto e^{(1-2\beta)a_4} = a^{-3}$: the condensate dilutes like dust. Its ratio is $w = x/(1+x)$ with $x = \lambda S/(2m) \propto a^{-3}$. Tuned to $w_0 = -1$ ($\lambda < 0$), it has $w_a = -6$, and its energy density $mS(1+x)$ vanishes at $z = 0.26$ and is negative before. For $w_0 = -0.75$, $w_a = -3.9$ and the zero is at $z = 0.33$. For every $w_0 \leq -0.3$ the zero lies below $z = 1$, where

the dark-energy density is measured to be positive. A $\lambda < 0$ mean field is moreover unbounded below.

- The vacuum energy (check E3). A natural Dirac-sea (Casimir) energy of Ψ on the cell, $m_{KK}^4/(16\pi^2) = 1.4 \times 10^{18}$ GeV^4, exceeds the dark-energy density 2.5×10^{-47} GeV^4 by 6×10^{64}. Calling it the dark energy restates the cosmological-constant problem; it does not solve it.
- What UM-1 uses: $\delta\Lambda/H^2 = 7.5 \times 10^{-93}$, a tuned geometric cosmological constant with $w = -1$ exactly. The record's observer on the primordial history sees $w_{exp} = -1$ for every b_4 (TAKEN): that acceleration is geometric, not Ψ.

Verdict: NOT SUPPORTED. Nothing in the dirac16complex sector of UM-1 behaves as dark energy. The association should not be made.

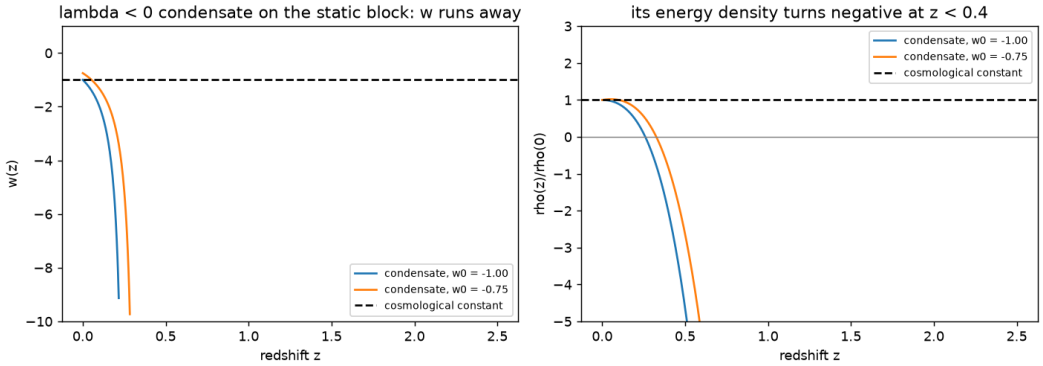


Figure 8: The lambda < 0 condensate on the static block: w runs away and the energy density turns negative at z < 0.4.

10. What this changes in UM-1

- `MODEL.md` section 1 said that "the uncertainty relation is a lower bound that every such eigenvalue satisfies". That is false for the zero modes and the light particles. Corrected there (section 1, section 4.4) to the statements of sections 5 and 6, with a new section 9 summarising this analysis.
- `FIELD_EQUATIONS.md` check D3 said "the mass squared of a bosonic mode is its mean squared hidden momentum". In the warped cell the Rayleigh quotient is the weighted mean of $|\partial_s u|^2$, the dressed momentum of section 3.1, not $\langle \hat{p}^2 \rangle$. The Gaussian identity of D3 holds on a flat line; the text is corrected.
- `SOLUTION.md` section 5.2 quoted the Standard-Model relic value 4.4×10^{-26} cm^3/s. UM-1 needs 6.5×10^{-26} (section 8.4). Corrected, with the unitarity fraction.
- `solution/masses.py`: the docstring's parametrisation of β (section 3.3) is recorded; the fit is unchanged.
- New requirements of the model, now explicit: the fold parity must be exact (section 8.3); the portal coefficients of the fold band and of the dark matter must differ, or the one-half count must hold (section 8.5); the primordial deflation must end before nucleosynthesis (section 7).

- The gaps these open, to be addressed in the next step: the origin of the light masses beyond the tip parametrisation; the derivation of the annihilation and portal coefficients; the dark-energy sector, which is $\delta\Lambda$.

11. Reproduction

From the repository root (each under `timeout -k 5 423`; the scripts locate their inputs from their own path):

```
cd Revision/universe_model
python analysis/x0_mass_analysis.py   # about 30 s, 21 checks
python analysis/dark_validity.py      # about 3 s, 13 checks; section 12
python tools/build_pdfs.py ANALYSIS
```

Inputs: `solution/outputs/masses.json`, `late_time.json`, `cosmology.json`, `ks/spectrum.json`, `ks/dm_lam0.json`, `ks/dm_lam1.json` (all from the solution, unchanged). Outputs: `analysis/outputs/x0_mass.json`, `analysis/outputs/dark_validity.json`, `analysis/outputs/generated_tables.md`, `analysis/reports/*.json`, `analysis/figures/fig*.png`. Gate steps: um1-analysis-x0 and um1-analysis-dark in `Revision/verify_revision.{sh,ps1}`.

12. Results

12.1 The mass decomposition and the Heisenberg comparison (`analysis/outputs/x0_mass.json`)

Units $(fL)^{-2}$ for the terms; Δs on one full period of x_0 (both halves); $m_H = \hbar c/(2\Delta s)$.

particle	$(mfL)^2$	$\langle p^2 \rangle$	**warp or** c^2	**fold**	**tip**
photon, gluon, graviton	0	20.25	20.25	-40.5	0.004999
W	1.325×10^{-5}	20.25	20.25	-40.5	0.004999
Z	1.705×10^{-5}	20.25	20.25	-40.5	0.004999
first even KK mode	30.12	9.87	20.25	-2.949	2.949
first odd mode	27.09	4.891	20.25	0	1.948
e (zero mode)	0	125.4	125.4	-250.7	4.721×10^{-8}
u (zero mode)	0	109	109	-218	1.861×10^{-7}
d (zero mode)	0	100.6	100.6	-201.3	3.891×10^{-7}
ν (zero mode)	0	382	382	-764	8.064×10^{-15}
Ψ fold band	0	1.125	1.125	-2.556	0.3064

Ψ dark matter (odd)	5.322	2.802	1.125	0.6974	0.6974
Ψ even bulk	10.99	9.87	1.125	-0.9521	0.9521

particle	Δs **[m]**	m **[GeV]**	m_H **[GeV]**	m/m_H	$\langle p^2\rangle \Delta s^2$
photon, gluon, graviton	1.40×10^{-21}	0	7.048×10^4	0	0.497
W	1.40×10^{-21}	80.37	7.048×10^4	0.00114	0.497
Z	1.40×10^{-21}	91.17	7.048×10^4	0.00129	0.497
first even KK mode	6.10×10^{-21}	1.212×10^5	1.619×10^4	7.49	4.59
first odd mode	5.56×10^{-21}	1.149×10^5	1.775×10^4	6.47	1.89
e (zero mode)	5.64×10^{-22}	5.110×10^{-4}	1.748×10^5	2.92×10^{-9}	0.5
u (zero mode)	6.05×10^{-22}	0.00216	1.630×10^5	1.32×10^{-8}	0.5
d (zero mode)	6.30×10^{-22}	0.0047	1.566×10^5	3.00×10^{-8}	0.5
ν (zero mode)	3.23×10^{-22}	5.000×10^{-11}	3.052×10^5	1.64×10^{-16}	0.5
Ψ fold band	3.79×10^{-21}	0	2.604×10^4	0	0.202
Ψ dark matter (odd)	5.06×10^{-21}	5.094×10^4	1.951×10^4	2.61	0.897
Ψ even bulk	5.48×10^{-21}	7.322×10^4	1.800×10^4	4.07	3.71

The fermion rows of the first table give the decomposition of the zero mode itself ($(mfL)^2 = 0$); the masses in the m column of the second come from the tip Yukawa coupling, $m_f = m_{KK}\zeta(c_f)^2$.

12.2 Sensitivity to the block size, $d\ln X/d\ln f$

X	$d\ln X/d\ln f$
Kaluza-Klein masses, the gap $3H/(2f)$	-1
m_W	-0.5007
m_Z	-0.5009
m_e	-22.39
m_u	-20.88
m_d	-20.06
m_ν	-39.09
M_{DM}	-0.6575
α	-4
G_N	-4

Λ_{QCD}, m_p (one-loop ESTIMATE)	-51.8
m_p/m_e	-29.41

Bound today (atomic clocks, FROM MEMORY): $|h_4|/H_0 < 1.2\times10^{-8}$; UM-1: $h_4/A = -8.86\times10^{-94}$.

12.3 Dark matter: the numbers (`analysis/outputs/dark_validity.json`)

quantity	**value**
M_{DM}	50.94 TeV
w of the $k = 0$ quanta, $\lambda = 0$ / λ_1	0 / 0.001817
$w_{eff}(A)$ on the primordial history	-0.07306
hidden pressure p_0/ρ	0.6575
largest allowed parity-violating amplitude	1.8×10^{-27}
freeze-out x_f, g_*	29.46, 60.25
needed $\langle\sigma v\rangle$	6.47×10^{-26} cm^3/s
thermal s-wave unitarity limit	1.73×10^{-25} cm^3/s (fraction used 0.374)
one-channel contact scale needed, $G^{-1/2}$	19.6 TeV ($G = 1.42\ /H^2$)
fold-band portal for $N_{eff} = 2.99$, $G^{-1/2}$	12.3 TeV
overlap ratio dark matter / fold band	0.329
dark-matter portal by the overlap rule, $G^{-1/2}$	21.5 TeV
$\langle\sigma v\rangle$ through the portal / needed	5.2 ($\Omega h^2 = 0.0231$)
σ_{SI} (overlap rule)	4.58×10^{-45} cm^2 (1.63 of the limit)
LZ limit at M_{DM}, linear extrapolation	2.80×10^{-45} cm^2
relic through the portal alone: $G^{-1/2}$, σ_{SI}	32.5 TeV, 8.80×10^{-46} cm^2
self-interaction σ/M for $G = 1/H^2$	2.9×10^{-18} cm^2/g

12.4 Dark energy: the $\lambda < 0$ condensate on the static block

w_0	w_a **(tangent)**	**energy density zero at** z
-1.2	-7.92	0.2239
-1	-6	0.2599
-0.9	-5.13	0.2828
-0.75	-3.938	0.3264
-0.5	-2.25	0.4422
-0.3	-1.17	0.6303

-0.1	-0.33	1.224

Natural vacuum energy of Ψ on the cell $m_{KK}^4/(16\pi^2) = 1.37 \times 10^{18}$ GeV^4; dark-energy density 2.47×10^{-47} GeV^4; ratio 5.5×10^{64}.